PhishGuard

Detect. Understand. Stay Safe.

PHISHING AWARENESS ANALYZER

Mini Project Documentation
A web-based cybersecurity awareness tool that highlights common phishing warning signs and teaches safer online practices.

Live Application
https://divergent-phish-guard-pro.base44.app

Prepared for: B.Tech Computer Science and Engineering
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Developed with Base44 AI App Builder

1. Abstract

PhishGuard is a web-based cybersecurity awareness project designed to help users recognize common warning signs in suspicious emails, SMS messages, and URLs. The application presents educational, rule-based analysis, explains why certain patterns may be risky, recommends safer actions, and includes an awareness quiz with fictional scenarios. The project emphasizes transparency: a warning is an indicator for the user to investigate, not a definitive verdict that a message or website is malicious.

The application was created using Base44, an AI-assisted application-building platform. The project demonstrates how a natural-language specification can be used to generate and refine a web application. Its intended role is educational awareness rather than replacing professional security products or live threat-intelligence services.

Keywords: Phishing awareness, cybersecurity, rule-based analysis, URL inspection, web application, Base44, security education.

2. Introduction

Phishing is a form of social engineering in which an attacker attempts to trick a person into revealing sensitive information, clicking a harmful link, opening an unsafe attachment, or making an unauthorized payment. Messages may create urgency, imitate trusted organizations, promise unexpected rewards, or request passwords and one-time passwords (OTPs).

Users can benefit from tools that explain these warning signs in plain language. PhishGuard combines simple analysis with learning material so that users can practice identifying suspicious patterns before interacting with real messages or links.

3. Problem Statement

Many users have difficulty distinguishing legitimate digital communications from phishing attempts. Warning signs can be subtle, and users may act quickly when a message creates fear or urgency. There is a need for an accessible educational tool that examines sample messages and visible URL characteristics, explains potential risks, and provides practical safety recommendations.

4. Objectives

- Help users learn common phishing indicators in emails, SMS messages, and links.
- Analyze user-provided sample text using transparent, explainable rules.
- Inspect visible URL characteristics and explain why they may deserve caution.
- Present an illustrative risk level with the detected reasons, rather than an unexplained score.
- Provide cybersecurity learning content and an interactive multiple-choice quiz.
- Demonstrate a practical web application created with an AI-assisted no-code/low-code tool.

5. Scope of the Project

PhishGuard focuses on awareness and basic pattern inspection. It is intended for learning, demonstrations, and practicing safe decision-making with fictional or non-sensitive examples.

Included in scope

- Message analysis for common warning phrases and requests for sensitive information.
- Visible URL checks such as unusual host formatting, IP-address hosts, URL shorteners, and HTTP usage.
- Explanations of detected indicators and suggested safe actions.
- Cybersecurity learning content and quiz questions with answer explanations.

Outside the current scope

- Definitive malware or phishing detection.
- Live reputation checks against external threat-intelligence databases.
- Opening, crawling, or executing submitted URLs.
- Guaranteeing that a message or website is safe.
- Replacing enterprise security tools, antivirus products, or professional incident response.

6. Proposed System

The proposed system accepts a sample message or URL, validates the input, applies predefined checks, and displays detected warning indicators with explanations. A separate learning module provides educational content and quiz-based practice.

Figure 1. High-level PhishGuard analysis workflow.

7. Main Modules

Module	Purpose
Home / Dashboard	Introduces PhishGuard, provides navigation to the analyzer and quiz, and summarizes common phishing warning signs.
Message Analyzer	Accepts an email or SMS sample and highlights patterns such as urgency, credential requests, payment demands, unexpected rewards, or suspicious links.
URL Warning Checker	Inspects the visible URL structure and reports characteristics that may warrant caution.
Learn Cyber Safety	Explains phishing, social engineering, suspicious attachments, OTP scams, and safer browsing habits.
Awareness Quiz	Uses fictional multiple-choice scenarios to help users practice identifying warning signs and learn from explanations.

8. Technology and Tools

Technology / Tool	Role in the project
Base44 AI App Builder	Used to create and refine the web application from natural-language instructions; the platform supports application UI, logic, and publishing workflows.
Web interface	Provides the pages, forms, buttons, result cards, learning content, and quiz interactions.
Rule-based analysis	Applies predefined patterns to submitted text and visible URL characteristics.
Browser	Used by the user to access and interact with the published website.
GitHub (documentation/repository)	Can be used to publish project documentation, README files, screenshots, and related project materials.

Implementation note

The project was built using Base44 as the AI-assisted development tool. This documentation describes the intended system and the publicly visible project description. Specific internal implementation details, such as generated source code, database configuration, and exact rule thresholds, should be documented only after confirming them in the Base44 project editor.

9. Rule-Based Analysis Approach

Rule-based analysis uses explicit conditions rather than an opaque model. Each check looks for a known characteristic and returns a warning label and explanation. This makes the results easier to understand and suitable for a cybersecurity-awareness demonstration.

Possible message indicators

- Urgency or threats: language that pressures the recipient to act immediately.
- Credential requests: requests for passwords, OTPs, or account verification.
- Payment pressure: unexpected requests to transfer money or pay a fee.
- Unexpected prizes: claims of rewards or winnings that the user did not enter.
- Suspicious links or sender mismatch: links or sender details that appear inconsistent or unusual.

Possible URL indicators

- HTTP instead of HTTPS, which means the connection is not protected by HTTPS; this alone does not prove phishing.
- An IP-address host or unusual hostname structure.
- URL shorteners that conceal the final destination.
- Look-alike spellings or misleading subdomains where the pattern is detectable.
- Malformed or unusually complex URLs that merit closer inspection.

Risk interpretation

A Low, Medium, or High label should be treated as an illustrative warning level based on the rules triggered. A low-risk result means that fewer configured warning signs were found; it does not establish that the message or URL is safe. False positives and false negatives are possible.

10. System Workflow

- The user opens the published PhishGuard website.
- The user chooses Message Analyzer, URL Warning Checker, or Awareness Quiz.
- For analysis, the user enters a fictional or non-sensitive message or URL.
- The application validates the input and applies its configured rule checks.
- The results page presents detected indicators, explanations, and recommended actions.
- For the quiz, the user answers questions and receives feedback and a final score.

11. Example Scenarios for Demonstration

The following examples are illustrative test scenarios. They are not a claim that every case has been executed or that the tool has a measured detection accuracy.

Scenario	Expected educational observation
Urgent message asking the user to verify a password through a link	May trigger urgency, credential-request, and link-related warnings.
Routine message about a lab assignment due tomorrow, with no sensitive request	May produce few or no warning indicators; this does not prove the message is legitimate.
URL containing an IP-address host	May trigger a URL-structure warning and recommendation to verify the destination independently.
Shortened URL from an unknown sender	May trigger a shortener warning because the final destination is not obvious from the displayed link.
Empty input	Should prompt the user to enter text or a URL before analysis.

12. Testing Plan

Test	Action	Expected result
T01	Submit empty message	Validation prompt; no analysis result.
T02	Submit fictional urgent credential request	Relevant indicators and explanations appear if matching rules are configured.
T03	Submit ordinary sample text	Only applicable indicators appear; no guarantee of safety.
T04	Submit malformed URL	URL validation message or clear format warning.
T05	Submit URL with visible suspicious structure	Applicable URL characteristics are explained.
T06	Answer quiz questions	Answers and final score are calculated consistently.
T07	Clear the analyzer	Input and prior result are reset.
T08	Use the site on a mobile screen	Layout remains usable without horizontal overflow.

Testing status: Execute these cases on the live application and update this section with Pass/Fail observations before submitting the final academic report.

13. Advantages

- Provides an approachable introduction to phishing awareness.
- Explains the reasons behind warnings instead of presenting only a score.
- Combines message analysis, URL inspection, learning material, and quizzes in one website.
- Rule-based checks are transparent and repeatable.
- Base44 reduces the amount of manual setup needed to generate and publish a web application.

14. Limitations

- Simple rules may miss sophisticated phishing attempts or flag legitimate messages.
- The tool does not confirm domain ownership, inspect website content, or query live threat-intelligence feeds unless those capabilities are explicitly added.
- HTTPS does not guarantee that a website is trustworthy.
- A suspicious characteristic is not proof of malicious intent, and no detected warning is not proof of safety.
- The quality of the educational analysis depends on the configured rules and the test cases used.

15. Future Enhancements

- Add an optional trusted threat-intelligence service after reviewing privacy, security, cost, and API requirements.
- Evaluate a machine-learning classifier using a suitable, labeled dataset and report its measured performance honestly.
- Add multilingual awareness content, including regional-language explanations.
- Expand test coverage with more legitimate and phishing-style examples.
- Add accessible design improvements and a structured feedback mechanism.

16. Conclusion

PhishGuard is an educational cybersecurity web application intended to help users identify common phishing warning signs, understand why a message or URL may deserve caution, and practice safer online behavior. By combining transparent rule-based analysis with learning content and quizzes, the project demonstrates practical web application design and cybersecurity awareness concepts. It should be used as a learning aid, not as a definitive security detector.

Project slogan: “Detect. Understand. Stay Safe.”

17. Live Application

Project URL: https://divergent-phish-guard-pro.base44.app

Development platform: Base44 AI App Builder

18. References

- PhishGuard live application: https://divergent-phish-guard-pro.base44.app
- Base44 AI App Builder: https://base44.com/ai-app-builder
- Base44 Features: https://base44.com/features

Prepared as a student mini-project document. Before publishing, fill in the student name, college, department details, and academic year on the cover page. Add screenshots of the actual application and update the test results after executing the test plan.